Mean, Variance, Standard deviation and Box plot

Q1. To compute the **mean (average)** of the given age data:

Step 1: Add all the values together

24+25+26+23+24+22+27+24=195

Step 2: Divide by the total number of values

There are **8** data points.

$$\text{Mean} = \frac{195}{8} = 24.375$$

✅ **Final Answer:**

Mean = 24.375

To compute the **variance** of the data:
Data: 24, 25, 26, 23, 24, 22, 27, 24
Mean (from before): 24.375

Step 1: Find the squared difference from the mean for each value:

$$(24 - 24.375)^2 = (-0.375)^2 = 0.1406$$
$$(25 - 24.375)^2 = (0.625)^2 = 0.3906$$
$$(26 - 24.375)^2 = (1.625)^2 = 2.6406$$
$$(23 - 24.375)^2 = (-1.375)^2 = 1.8906$$
$$(24 - 24.375)^2 = (-0.375)^2 = 0.1406$$
$$(22 - 24.375)^2 = (-2.375)^2 = 5.6406$$
$$(27 - 24.375)^2 = (2.625)^2 = 6.8906$$
$$(24 - 24.375)^2 = (-0.375)^2 = 0.1406$$

Step 2: Sum of squared differences:

0.1406+0.3906+2.6406+1.8906+0.1406+5.6406+6.8906+0.1406=17.8748

Step 3: Divide by n or n−1

- For **population variance**, divide by **n = 8**

$$\text{Population Variance} = \frac{17.8748}{8} = 2.23435$$

- For **sample variance**, divide by **n-1 = 7**

$$\text{Sample Variance} = \frac{17.8748}{7} = 2.5535$$

✅ **Final Answer:**

- **Population variance = 2.23** (rounded to 2 decimal places)
- **Sample variance = 2.55** (rounded to 2 decimal places)

Q2. Let's compute the **mean** of the following data:

Data:

24, 25, 26, 23, 24, 22, 27, 24, 60

Step 1: Add all values together

24+25+26+23+24+22+27+24+60=255

Step 2: Count the number of values

There are **9** data points.

Step 3: Compute the mean

$$\text{Mean} = \frac{255}{9} \approx 28.33$$

✅ **Final Answer:**

Mean = 28.33 (rounded to 2 decimal places)

Let's compute the **variance** for the data:

Data:

24, 25, 26, 23, 24, 22, 27, 24, 60
Mean = 28.33 (from previous calculation)

Step 1: Compute squared differences from the mean

$$(24 - 28.33)^2 = (-4.33)^2 = 18.7489$$
$$(25 - 28.33)^2 = (-3.33)^2 = 11.0889$$
$$(26 - 28.33)^2 = (-2.33)^2 = 5.4289$$
$$(23 - 28.33)^2 = (-5.33)^2 = 28.4089$$
$$(24 - 28.33)^2 = (-4.33)^2 = 18.7489$$
$$(22 - 28.33)^2 = (-6.33)^2 = 40.0689$$
$$(27 - 28.33)^2 = (-1.33)^2 = 1.7689$$
$$(24 - 28.33)^2 = (-4.33)^2 = 18.7489$$
$$(60 - 28.33)^2 = (31.67)^2 = 1002.5889$$

Step 2: Sum of squared differences

$$\text{Sum} = 18.7489 + 11.0889 + 5.4289 + 28.4089 + 18.7489 + 40.0689 + 1.7689 + 18.7489 + 1002.5889 = 1145.6001$$

Step 3: Divide by n (population) or n - 1 (sample)

- **Population variance**:

$$\frac{1145.6001}{9} \approx 127.29$$

- **Sample variance**:

$$\frac{1145.6001}{8} \approx 143.20$$

✅ Final Answer:

- **Population variance ≈ 127.29**
- **Sample variance ≈ 143.20**

Let's interpret the **variance** and **standard deviation** using your age data:

Data:

24, 25, 26, 23, 24, 22, 27, 24, 60
Mean = 28.33
Sample variance ≈ 143.20
Sample standard deviation = √143.20 ≈ 11.96

🔍 What does variance mean in this context?

- **Variance** measures how spread out the ages are from the mean (average).
- A **higher variance** means the values are **more spread out**.
- In this case, the variance is **143.20**, which is relatively high **because of the outlier: 60**.
- Most of the ages are in the 22–27 range, but 60 is far from the mean and increases the variance significantly.

📏 What does standard deviation mean?

- The **standard deviation** is the square root of the variance and gives a measure of spread **in the same unit** as the data (here, **years**).
- A standard deviation of ≈ **11.96 years** means that, on average, the ages deviate about 12 years from the mean of 28.33.
- Again, this large value mostly results from the single outlier (60).

📌 Interpretation Summary:

- Most of the ages are around 24–27 years, but the **presence of the outlier (60)** causes the **mean to increase** and the **variance and standard deviation to be large**.
- If that outlier were removed, you'd likely see a **much lower variance and standard deviation**, indicating that the ages are more closely grouped around the mean.

Q. Let's **draw a box plot** for the data:
Data: 24, 25, 26, 23, 24, 22, 27, 24, 60

Step 1: Sort the data

Sorted data:
22, 23, 24, 24, 24, 25, 26, 27, 60

Step 2: Find the five-number summary

1. **Minimum:** 22
2. **Q1 (1st Quartile):** Median of lower half = median of (22, 23, 24, 24)

$$Q1 = \frac{23 + 24}{2} = 23.5$$

3. **Median (Q2):** Middle value = **24**
4. **Q3 (3rd Quartile):** Median of upper half = median of (25, 26, 27, 60)

$$Q3 = \frac{26 + 27}{2} = 26.5$$

5. **Maximum:** 60

Step 3: Calculate the Interquartile Range (IQR)

IQR=Q3−Q1 = 26.5 - 23.5 = 3

Step 4: Determine outlier boundaries

- **Lower Fence/bound** = Q1 – 1.5 × IQR = 23.5 – 4.5 = **19**
- **Upper Fence/bound** = Q3 + 1.5 × IQR = 26.5 + 4.5 = **31**

Any data **above 31 or below 19** is considered an outlier.
✅ So, **60 is an outlier**.

Step 5: Draw the box plot

I'll represent it **textually** here:

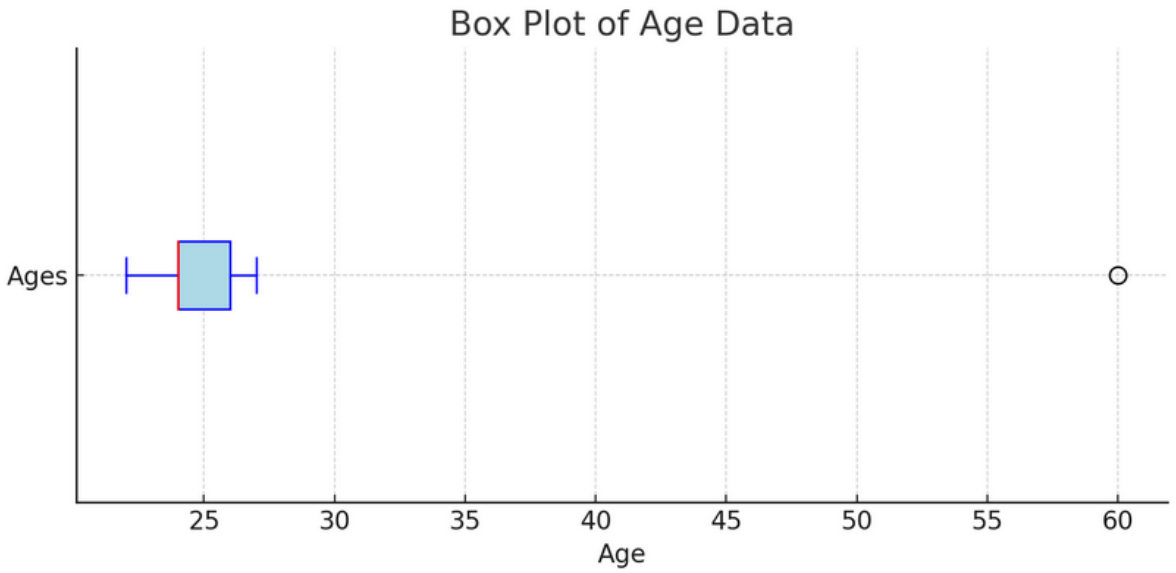
Outlier: *

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        |
        |
22  23.5  24    26.5    27    60
 |----|=====|=======|-----|    *
    Q1   Median  Q3
```

- Box spans from **Q1 (23.5)** to **Q3 (26.5)**
- **Median = 24** is inside the box
- **Whiskers** extend from **Min = 22** to **Max before outlier = 27**
- **60** is shown as an **outlier**

Here is the box plot for your age data. You can clearly see:

- The **box** spans from Q1 (23.5) to Q3 (26.5),
- The **red line** shows the median (24),
- The **whiskers** extend to the minimum (22) and to the last non-outlier (27),
- The **dot at 60** is an **outlier**.



Let's distinguish **empirical formulas** from **other (theoretical or exact) mathematical formulas**:

🔍 Empirical Formula vs Theoretical Formula

Feature	Empirical Formula	Theoretical/Mathematical Formula
Definition	Based on **observed data or experiments**	Based on **mathematical derivation or laws**
Accuracy	**Approximate**, works well in practice	**Exact**, under defined conditions
Origin	Discovered by **fitting data** (data-driven)	Derived using **logic, math, or physical laws**
Example (Statistics)	Empirical Rule: 68-95-99.7 rule for normal distribution	Standard deviation formula: $\sqrt{\frac{1}{n}\sum(x_i - \bar{x})^2}$
Example (Physics)	Newton's cooling law (from experiment)	F = ma (from Newton's 2nd law — theoretical)
When used	When theory is too complex or unknown	When underlying theory is well understood
Flexibility	Can change with new data	Fixed — valid as long as assumptions hold

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✅ In Simple Terms:

- **Empirical formulas**:
  → **"We observed this pattern in the data"**
  → Example: "Most people's heights are within 2 SDs of the mean"
- **Theoretical formulas**:
  → **"We proved this using laws or equations"**
  → Example: Area of circle = πr^2 — no data needed, it's proven

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📌 When do we use empirical formulas?

- When it's hard or impossible to derive a formula from first principles.
- When you want a **quick approximation** (e.g., engineering, biology, machine learning).
- When working with **real-world messy data** (like weather forecasting).

Certainly! Here's a **structured and in-depth interpretation** of the two datasets with the **same mean (80.0)** but **different variances**, suitable for academic or analytical reporting.

🎓 Context: Student Exam Score Analysis

Here are two datasets with **exactly the same mean (80.0)** but **very different variances**, shown in the table above.

Student	Score_A (Low Variance)	Score_B (High Variance)
Student 1	78	60
Student 2	79	70
Student 3	80	80
Student 4	81	90
Student 5	82	100

📊 Statistical Summary

Dataset	Mean	Sample Variance	Sample Standard Deviation
A (Low Variance)	80.0	2.50	1.58
B (High Variance)	80.0	250.00	15.81

We analyzed test scores from two different classes (Dataset A and Dataset B), each with 5 students. While both classes have the **same average score of 80**, the **spread of the scores differs significantly**.

📊 Summary Statistics

Metric	Dataset A (Low Variance)	Dataset B (High Variance)
Scores	78, 79, 80, 81, 82	60, 70, 80, 90, 100
Sample Mean ($\bar{x}$)	80.0	80.0
Sample Variance (s^2)	2.50	250.00
Sample Std. Deviation (s)	1.58	15.81

🔍 Interpretation

1. Central Tendency (Mean)

- Both datasets have an **identical mean of 80.0**, indicating that the **average performance** in each class is the same.
- However, relying on the mean alone **masks the true nature** of student performance in each group.

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2. Dispersion (Variance & Standard Deviation)

Aspect	Dataset A	Dataset B
Variance (s^2)	2.5	250.0
Standard Deviation (s)	1.58	15.81
Spread of Scores	Narrow	Wide

- **Dataset A**:

  - ✓ Scores are **clustered tightly around the mean**.
  - ✓ Reflects **consistent performance**.
  - ✓ Suggests the class is relatively **homogeneous** in preparation and understanding.
  - ✓ Indicates the **test may have been fair** and within most students' capabilities.

- **Dataset B**:

  - ✓ Scores vary widely, from **60 to 100**.
  - ✓ Reflects **heterogeneous performance**.
  - ✓ Indicates **disparities in learning**, possibly due to:

    - Unequal access to learning resources
    - Varying levels of understanding or effort
    - Potential external factors affecting individual students

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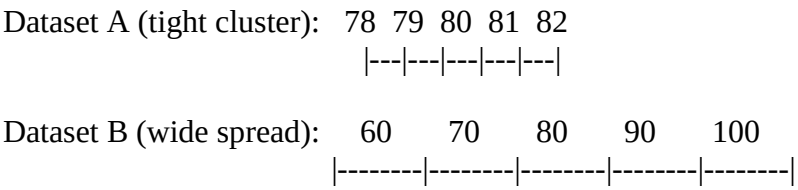
3. Educational Implications

Dimension	Dataset A	Dataset B
Instruction Effectiveness	Likely effective for all	May need differentiation
Intervention Need	Low	High (especially for low scorers)
Evaluation	Uniform outcomes → fair test	Diverse outcomes → mixed understanding

- **Dataset A** would be seen as an academically balanced class.
- **Dataset B** would prompt a deeper investigation into:
  - ✓ Why some students are underperforming
  - ✓ Whether the curriculum or instruction needs differentiation
  - ✓ If assessments are biased toward or against certain students

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4. Visual Insight (Mental Picture)



✅ Conclusion

Although **mean scores are equal**, the **variance tells a deeper story**.
Dataset A suggests consistent student performance, while **Dataset B** reveals a wide range of abilities or external influences.

📌 **Key Insight**: Always pair measures of central tendency with measures of dispersion for a **complete understanding** of data.
